BBD JavaScript Coding Practice

9 practice questions • Easy, Medium & Hard

A self-practice worksheet for your coding assessment preparation.

Instructions

- Attempt the questions in order, starting with Easy and progressing to Hard.
- Use JavaScript and write a function for each problem.
- Test your solutions with the examples and think about edge cases.
- Unless stated otherwise, avoid relying on built-in methods that the question prohibits.
- No solutions or answer key are included, so you can practise independently.

Name: _____ Date: _____

EASY — 3 QUESTIONS

1. Count Even Numbers

Write a function that receives an array of integers and returns how many numbers are even.

```
function countEvenNumbers(numbers) {
    // your code
}

console.log(countEvenNumbers([1, 2, 3, 4, 6])); // 3
console.log(countEvenNumbers([1, 3, 5]));       // 0
```

Concepts: arrays, loops, conditionals.

2. Reverse a String

Write a function that reverses a string without using .reverse().

```
function reverseString(str) {
    // your code
}

console.log(reverseString("hello")); // "olleh"
console.log(reverseString("BBD"));   // "DBB"
```

Concepts: strings, loops, indexing.

3. Find the Maximum

Write a function that returns the largest number in an array. Do not use Math.max().

```
function findMax(numbers) {
    // your code
}

console.log(findMax([4, 7, 2, 9, 1])); // 9
console.log(findMax([-5, -2, -10]));    // -2
```

Concepts: arrays, comparison, iteration.

MEDIUM — 3 QUESTIONS

4. Two Sum

Given an array of integers and a target number, return the indices of two numbers whose sum equals the target. Assume there is exactly one solution.

```
function twoSum(numbers, target) {
    // your code
}

console.log(twoSum([2, 7, 11, 15], 9)); // [0, 1]
console.log(twoSum([3, 2, 4], 6));       // [1, 2]
```

Challenge: aim for O(n) time. Concepts: arrays, Map/object lookup, searching, complexity.

5. First Non-Repeating Character

Return the first character that occurs only once. Return null if every character repeats.

```
function firstUnique(str) {
    // your code
}

console.log(firstUnique("aabbcdd")); // "c"
console.log(firstUnique("swiss"));   // "w"
console.log(firstUnique("aabb"));    // null
```

Concepts: strings, frequency counting, objects/Map.

6. Remove Duplicates

Given an array, return a new array containing each value only once, preserving the original order. Do not use Set.

```
function removeDuplicates(numbers) {
    // your code
}

console.log(removeDuplicates([1, 2, 2, 3, 1, 4])); // [1, 2, 3, 4]
console.log(removeDuplicates(["a", "b", "a", "c", "b"])); // ["a", "b", "c"]
```

Concepts: arrays, objects, searching.

HARD — 3 QUESTIONS

7. Longest Consecutive Sequence

Given an unsorted array, find the length of the longest sequence of consecutive integers. The numbers need not be adjacent in the input. Do not sort the array. Duplicates and negative numbers may occur; an empty array returns 0.

```
function longestConsecutive(numbers) {
    // your code
}

console.log(longestConsecutive([100, 4, 200, 1, 3, 2])); // 4
console.log(longestConsecutive([0, 3, 7, 2, 5, 8, 4, 6, 0, 1])); // 9
console.log(longestConsecutive([])); // 0
```

Target: O(n) time. Concepts: Set/lookup strategy, sequences, complexity.

8. Balanced Brackets

Given a string containing only the bracket characters (), {}, and [], determine whether the brackets are correctly balanced and nested.

```
function isBalanced(str) {
    // your code
}

console.log(isBalanced("()"));     // true
console.log(isBalanced("({[]})")); // true
console.log(isBalanced("([)]"));   // false
console.log(isBalanced("((("));    // false
console.log(isBalanced("{[()]}")); // true
```

Hint: think about a stack. Concepts: strings, arrays, stacks, algorithms.

9. Maximum Subarray Sum

Given an array containing positive and negative integers, find the largest possible sum of a non-empty contiguous subarray.

```
function maxSubarraySum(numbers) {
    // your code
}

console.log(maxSubarraySum([-2, 1, -3, 4, -1, 2, 1, -5, 4])); // 6
console.log(maxSubarraySum([5, -2, 3, 4])); // 10
```

Target: O(n) time. Concepts: arrays, dynamic programming, optimization.